

Education

San José State University, College of Engineering,

San Jose, CA, USA, Jan 2022 – May 2022

Exchange Student, Cumulative GPA: 3.925/4.00.

Cairo University, Faculty of Engineering,

Giza, Egypt, 2018-2023

B.S. Computer Engineering, Cumulative GPA: 3.26/4.00

Honors & Awards

- Higher Education Initiative Scholarship**, awarded a full scholarship for a B.S. in Computer Engineering with intensive English and soft skills trainings, culminating in receiving the Awesome Student Award in my final semester.

Work Experience

Avidbeam, Full-stack Developer

- Developed and maintained backend and frontend features using Ruby on Rails, React, Next, and MongoDB, enhancing user experience and application performance.
- Redesigned system architecture from a standalone cluster to a federated cluster-of-clusters model, where each cluster can act as both a parent and a child, enabling hierarchical data federation, centralized control, and decentralized scalability.
- Integrated Authentik as a centralized identity provider, replacing custom authentication to support inter-cluster authentication and external services like LDAP and OAuth.
- Migrated 50–70 million records efficiently during system upgrades with zero downtime and performance impact.
- Built a proto generator app, improving internal development workflows and boosting team efficiency.
- Ensured code reliability and maintainability through rigorous testing with RSpec and RSwag.
- Collaborated with cross-functional teams to troubleshoot critical issues for efficient application management.

Sep 2023 - Present
Cairo, Egypt

Cairo University, Faculty of Engineering, External Teaching Assistant

- Assisting students in understanding the concepts for the Introduction to Database Management Systems, and Software Engineering courses.
- Supporting faculty with grading, preparing materials, and providing guidance to students during labs and office hours.

Sep 2024 - Present
Giza, Egypt

Hoopoe Digital, Web Development Intern.

- Developed an analytics platform and indoor mapping application with interactive D3.js visualizations and user-friendly editing forms using Agile methodologies.
- Built forms for map admins to edit geofence styles, utilizing React.js, GitLab, and Trello.

Jul 2022 - Sep 2022
Cairo, Egypt

Bosta, Quality Control Intern.

- Tested and documented web and mobile applications and APIs in an Agile environment, documented test cases and results in Jira, and utilized Postman and 3T Studio

Jul 2022 - Sep 2022
Cairo, Egypt

Software Projects

Shopping Website,

Dec 2023 – Jan 2024

- Created an e-commerce platform with MERN stack; integrated Cloudinary for image handling.
- Developed dual user interfaces: admin for product and order management, and customer for browsing and purchasing products.
- Implemented features allowing customers to filter products by different factors, add products to their cart, complete purchases, track order history, cancel orders before delivery, and reset passwords using SendGrid.

Graduation Project: Opinion Mining Application, Cairo University

Sep 2022 – May 2023

- Designed and implemented an advanced Opinion Mining system for extracting insights, quotations, summaries and sentiment from diverse sources like news articles, blogs, and social media. Enabled users to quickly access relevant information about specific topics or individuals, improving their information-gathering process.
- Developed a user-friendly website for journalists and professionals to analyze and compare opinions on a given topic. Built a scalable system utilizing NLP techniques to extract meaningful insights from unstructured text, enabling informed decision-making. Implemented the website using React and Python, ensuring real-time data gathering and processing capabilities.

Job Listing Website, San Jose State University

Sep 2021 – Jan 2022

- Developed a comprehensive job listing platform to facilitate efficient job searches and employee recruitment.
- Implemented distinct user interfaces for employers and employees, enhancing user experience and functionality.
- Created key features including a profile page, job pages, a chat system using WebSockets for real-time communication, and an admin page for employers to review applications and communicate with candidates.
- Utilized ReactJS, NodeJS, ExpressJS, MongoDB, and WebSockets for full-stack development.
- Managed project using Agile methodologies with Zenhub for tracking progress and git for code management.

World Cup Match Reservation System, Cairo University

Dec 2022 Jan 2023

- Developed an interactive sports reservation website to display teams, schedules, and seating availability.
- Implemented functionalities for users to reserve seats, view their reservations, and cancel them as needed.
- Enabled managers to add and edit matches and stadiums, manage match details, and delete entries.
- Designed an administrative interface for approving manager registrations.
- Built with MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring a responsive and dynamic user experience.

OS scheduler, Cairo University

Nov 2021 – Jan 2022

- Developed process generator, schedulers (HPF, RR, SRTN), and buddy memory allocation, enhancing problem solving abilities.

Hospital Management application, Cairo University

Sep 2020 – Jan 2021

- Produced a Windows application using C# and SQL, including database schema design and database connectivity.

Castle Game, Applied OOP, Data structures, and problem-solving techniques in C++ to simulate castle fighting.

Dec 2021

Snake & ladder with monopoly cards Game, using OOP implemented the logic of the game in C++.

May 2020

Skills

Technical Skills: Programming (JavaScript, TypeScript, Ruby, C++, Python, Java, C, C#, SQL, MongoDB, Elasticsearch, Verilog, VHDL, MATLAB, HTML, CSS), **Web Technologies** (Next.js, React.js, Node.js, Bootstrap, jQuery, TailwindCSS, Express, Rails, Websockets, Chakra UI), **Tools** (Git, GitHub, GitLab, Jira, Trello, Zen Hub, Postman), **Others** (Agile, Microsoft Office, Word, PowerPoint, and Excel)

Soft Skills: Leadership, Problem Solving, Communication, Teamwork, Management.

Languages: Arabic (Native), English (B2)

Leadership & Activities

- **Head** | Stem for sustainability event startup pitching head.
 - **Organizer** | Sawiris scholarship contract agreement conference & (2022 – 2024) HEI’s Graduation ceremony
 - **Vice-head** | Logistics team SuccessHunter2020 Educational Summit. 2019 - 2024
 - **Initiative leader** | Community Service initiative teaching students Microsoft Office.
 - **Instructor** | AYP Technology team & Resala’s literacy project & English camp for school students
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